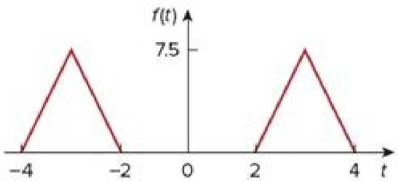
Chapter 18, Problem 5PP

Problem

Determine the Fourier transform of the function in Fig. 18.16.

Figure 18.16



Step-by-step solution

Step 1 of 3

Refer the Figure 18.16, in the text book.

The Fourier transform can be found by using derivative property as shown in figure 1.

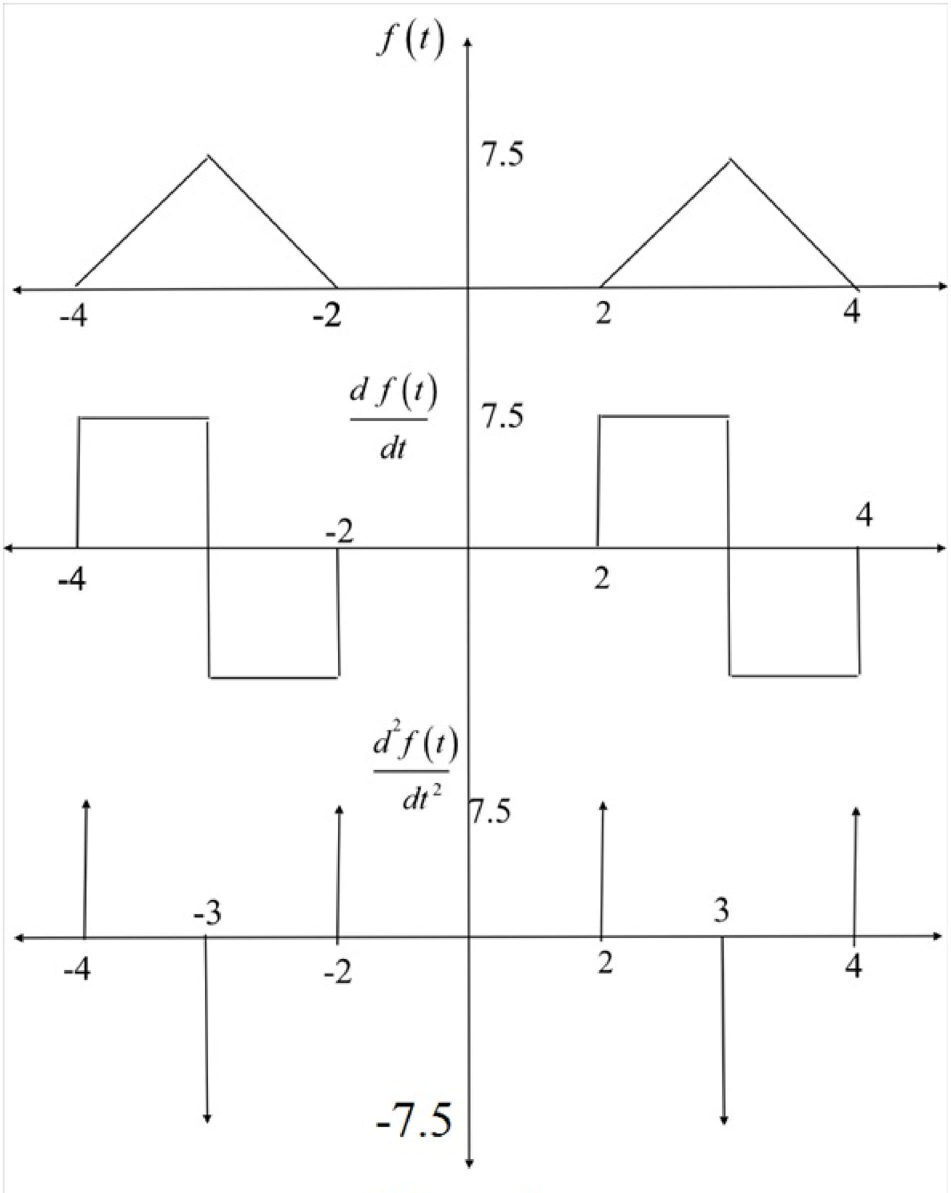


Figure 1

Step 2 of 3

Figure 1 shows the signals obtained by successive differentiation of $f(t)$.

The signal $\frac{d^2 f(t)}{dt^2}$ can be expressed as

$$\frac{d^2 f(t)}{dt^2}=7.5\left[\delta(t+4)-2\delta(t+3)+\delta(t+2)+\delta(t-2)-2\delta(t-3)+\delta(t-4)\right]$$

Using time differentiation property of Fourier transform,

$$F\left[\frac{d^2 f(t)}{dt^2}\right]=7.5F\left[\delta(t+4)-2\delta(t+3)+\delta(t+2)+\delta(t-2)-2\delta(t-3)+\delta(t-4)\right]$$

$$F\left(\frac{d^2 f(t)}{dt^2}\right)=(j\omega)^2 F(j\omega)$$

And $\delta(t\pm x)\leftrightarrow e^{\pm j\omega x}$.

$$\begin{aligned}(j\omega)^2 F(j\omega)&=7.5\left(e^{j4\omega}-2e^{j3\omega}+e^{j2\omega}+e^{-j2\omega}-2e^{-j3\omega}+e^{-j4\omega}\right)\\&=7.5\left[\left(e^{j4\omega}+e^{-j4\omega}\right)-2\left(e^{j3\omega}+e^{-j3\omega}\right)+\left(e^{j2\omega}+e^{-j2\omega}\right)\right]\\&=7.5\left[2\left(\frac{e^{j4\omega}+e^{-j4\omega}}{2}\right)-4\left(\frac{e^{j3\omega}+e^{-j3\omega}}{2}\right)+2\left(\frac{e^{j2\omega}+e^{-j2\omega}}{2}\right)\right]\end{aligned}$$

Step 3 of 3

Simplify further.

$$\begin{aligned}F(j\omega)&=\frac{7.5(2\cos 4\omega-4\cos 3\omega+2\cos 2\omega)}{(j\omega)^2}\\&=\frac{15\cos 4\omega-30\cos 3\omega+15\cos 2\omega}{-\omega^2}\\&=\frac{30\cos 3\omega-15\cos 4\omega-15\cos 2\omega}{\omega^2}\end{aligned}$$

Therefore, the Fourier transform of $f(t)$ is,

$$\boxed{F(j\omega)=\frac{30\cos 3\omega-15\cos 4\omega-15\cos 2\omega}{\omega^2}}.$$